

100+ Python challenging programming exercises for Python 3

1. Level description

Level 1 Beginner

Beginner means someone who has just gone through an introductory Python course. He can solve some problems with 1 or 2 Python classes or functions. Normally, the answers could directly be found in the textbooks.

Level 2 Intermediate

Intermediate means someone who has just learned Python, but already has a relatively strong programming background from before. He should be able to solve problems which may involve 3 or 3 Python classes or functions. The answers cannot be directly be found in the textbooks.

Level 3 Advanced.

He should use Python to solve more complex problem using more rich libraries functions and data structures and algorithms. He is supposed to solve the problem using several Python standard packages and advanced techniques.

2. Problem template

Question

Hints

Solution

3. Questions

Question 1

Level 1

Question:

Write a program which will find all such numbers which are divisible by 7 but are not a multiple of 5, between 2000 and 3200 (both included).

The numbers obtained should be printed in a comma-separated sequence on a single line.

Hints:

Consider use range(#begin, #end) method

Solution:

```
l=[]
for i in range(2000, 3201):
    if (i%7==0) and (i%5!=0):
        l.append(str(i))

print(','.join(l))
```

Question 2

Level 1

Question:

Write a program which can compute the factorial of a given numbers.

The results should be printed in a comma-separated sequence on a single line.

Suppose the following input is supplied to the program:

8

Then, the output should be:

40320

Hints:

In case of input data being supplied to the question, it should be assumed to be a console input.

Solution:

```
def fact(x):
    if x == 0:
        return 1
    return x * fact(x - 1)

x=int(input())
print(fact(x))
```

Question 3

Level 1

Question:

With a given integral number n , write a program to generate a dictionary that contains $(i, i*i)$ such that i is an integral number between 1 and n (both included). and then the program should print the dictionary.

Suppose the following input is supplied to the program:

8

Then, the output should be:

{1: 1, 2: 4, 3: 9, 4: 16, 5: 25, 6: 36, 7: 49, 8: 64}

Hints:

In case of input data being supplied to the question, it should be assumed to be a console input.

Consider use dict()

Solution:

```
n=int(input())
d=dict()
for i in range(1,n+1):
    d[i]=i*i

print(d)
```

Question 4

Level 1

Question:

Write a program which accepts a sequence of comma-separated numbers from console and generate a list and a tuple which contains every number.

Suppose the following input is supplied to the program:

34,67,55,33,12,98

Then, the output should be:

['34', '67', '55', '33', '12', '98']

('34', '67', '55', '33', '12', '98')

Hints:

In case of input data being supplied to the question, it should be assumed to be a console input.

tuple() method can convert list to tuple

Solution:

```
values=input()
l=values.split(",")
t=tuple(l)
print(l)
print(t)
```

Question 5

Level 1

Question:

Define a class which has at least two methods:

getString: to get a string from console input

printString: to print the string in upper case.

Also please include simple test function to test the class methods.

Hints:

Use **init** method to construct some parameters

Solution:

```
class InputOutString(object):
    def __init__(self):
        self.s = ""

    def getString(self):
        self.s = input()

    def printString(self):
        print(self.s.upper())

strObj = InputOutString()
strObj.getString()
strObj.printString()
```

Question 6

Level 2

Question:

Write a program that calculates and prints the value according to the given formula:

$Q = \text{Square root of } [(2 * C * D)/H]$

Following are the fixed values of C and H:

C is 50. H is 30.

D is the variable whose values should be input to your program in a comma-separated sequence.

Example

Let us assume the following comma separated input sequence is given to the program:

100,150,180

The output of the program should be:

18,22,24

Hints:

If the output received is in decimal form, it should be rounded off to its nearest value (for example, if the output received is 26.0, it should be printed as 26)

In case of input data being supplied to the question, it should be assumed to be a console input.

Solution:

```
import math
c=50
h=30
value = []
items=[x for x in input().split(',')]
for d in items:
    value.append(str(int(round(math.sqrt(2*c*float(d)/h))))))

print(','.join(value))
```

Question 7

Level 2

Question:

Write a program which takes 2 digits, X,Y as input and generates a 2-dimensional array. The element value in the i-th row and j-th column of the array should be $i*j$.

Note: $i=0,1.., X-1$; $j=0,1,iY-1$.

Example

Suppose the following inputs are given to the program:

3,5

Then, the output of the program should be:

[[0, 0, 0, 0, 0], [0, 1, 2, 3, 4], [0, 2, 4, 6, 8]]

Hints:

Note: In case of input data being supplied to the question, it should be assumed to be a console input in

a comma-separated form.

Solution:

```
input_str = input()
dimensions=[int(x) for x in input_str.split(',')]
rowNum=dimensions[0]
colNum=dimensions[1]
multilist = [[0 for col in range(colNum)] for row in range(rowNum)]

for row in range(rowNum):
    for col in range(colNum):
        multilist[row][col]= row*col

print(multilist)
```

Question 8

Level 2

Question:

Write a program that accepts a comma separated sequence of words as input and prints the words in a comma-separated sequence after sorting them alphabetically.

Suppose the following input is supplied to the program:

without,hello,bag,world

Then, the output should be:

bag,hello,without,world

Hints:

In case of input data being supplied to the question, it should be assumed to be a console input.

Solution:

```
items=[x for x in input().split(',')]
items.sort()
print(','.join(items))
```

Question 9

Level 2

Question 9

Write a program that accepts sequence of lines as input and prints the lines after making all characters in the sentence capitalized.

Suppose the following input is supplied to the program:

Hello world

Practice makes perfect

Then, the output should be:

HELLO WORLD

PRACTICE MAKES PERFECT

Hints:

In case of input data being supplied to the question, it should be assumed to be a console input.

Solution:

```
lines = []
while True:
    s = input()
    if s:
        lines.append(s.upper())
    else:
        break;

for sentence in lines:
    print(sentence)
```

Question 10

Level 2

Question:

Write a program that accepts a sequence of whitespace separated words as input and prints the words after removing all duplicate words and sorting them alphanumerically.

Suppose the following input is supplied to the program:

hello world and practice makes perfect and hello world again

Then, the output should be:

again and hello makes perfect practice world

Hints:

In case of input data being supplied to the question, it should be assumed to be a console input.

We use set container to remove duplicated data automatically and then use sorted() to sort the data.

Solution:

```
s = input()
words = [word for word in s.split(" ")]
print(" ".join(sorted(list(set(words)))))
```

Question 11

Level 2

Question:

Write a program which accepts a sequence of comma separated 4 digit binary numbers as its input and then check whether they are divisible by 5 or not. The numbers that are divisible by 5 are to be printed in a comma separated sequence.

Example:

0100,0011,1010,1001

Then the output should be:

1010

Notes: Assume the data is input by console.

Hints:

In case of input data being supplied to the question, it should be assumed to be a console input.

Solution:

```
value = []
items=[x for x in input().split(',') ]
for p in items:
    intp = int(p, 2)
    if not intp%5:
        value.append(p)

print(','.join(value))
```

Question 12

Level 2

Question:

Write a program, which will find all such numbers between 1000 and 3000 (both included) such that each digit of the number is an even number.

The numbers obtained should be printed in a comma-separated sequence on a single line.

Hints:

In case of input data being supplied to the question, it should be assumed to be a console input.

Solution:

```
values = []
for i in range(1000, 3001):
    s = str(i)
    if (int(s[0])%2==0) and (int(s[1])%2==0) and (int(s[2])%2==0) and (ir
        values.append(s)
print(",".join(values))
```

Question 13

Level 2

Question:

Write a program that accepts a sentence and calculate the number of letters and digits.

Suppose the following input is supplied to the program:

hello world! 123

Then, the output should be:

LETTERS 10

DIGITS 3

Hints:

In case of input data being supplied to the question, it should be assumed to be a console input.

Solution:

```
s = input()
d={"DIGITS":0, "LETTERS":0}
for c in s:
    if c.isdigit():
        d["DIGITS"]+=1
    elif c.isalpha():
        d["LETTERS"]+=1
    else:
        pass
print("LETTERS", d["LETTERS"])
print("DIGITS", d["DIGITS"])
```

Question 14

Level 2

Question:

Write a program that accepts a sentence and calculate the number of upper case letters and lower case letters.

Suppose the following input is supplied to the program:

Hello world!

Then, the output should be:

UPPER CASE 1

LOWER CASE 9

Hints:

In case of input data being supplied to the question, it should be assumed to be a console input.

Solution:

```
s = input()
d={"UPPER CASE":0, "LOWER CASE":0}
for c in s:
    if c.isupper():
        d["UPPER CASE"]+=1
    elif c.islower():
        d["LOWER CASE"]+=1
    else:
        pass
print("UPPER CASE", d["UPPER CASE"])
print("LOWER CASE", d["LOWER CASE"])
```

Question 15

Level 2

Question:

Write a program that computes the value of $a+aa+aaa+aaaa$ with a given digit as the value of a .

Suppose the following input is supplied to the program:

9

Then, the output should be:

11106

Hints:

In case of input data being supplied to the question, it should be assumed to be a console input.

Solution:

```

a = input()
n1 = int( "%s" % a )
n2 = int( "%s%s" % (a,a) )
n3 = int( "%s%s%s" % (a,a,a) )
n4 = int( "%s%s%s%s" % (a,a,a,a) )
print(n1+n2+n3+n4)

```

Question 16

Level 2

Question:

Use a list comprehension to square each odd number in a list. The list is input by a sequence of comma-separated numbers.

Suppose the following input is supplied to the program:

1,2,3,4,5,6,7,8,9

Then, the output should be:

1,3,5,7,9

Hints:

In case of input data being supplied to the question, it should be assumed to be a console input.

Solution:

```

values = input()
numbers = [x for x in values.split(",") if int(x)%2!=0]
print(",".join(numbers))

```

Question 17

Level 2

Question:

Write a program that computes the net amount of a bank account based a transaction log from console input. The transaction log format is shown as following:

D 100

W 200

D means deposit while W means withdrawal.

Suppose the following input is supplied to the program:

D 300

D 300

W 200

D 100

Then, the output should be:

500

Hints:

In case of input data being supplied to the question, it should be assumed to be a console input.

Solution:

```
netAmount = 0
while True:
    s = input()
    if not s:
        break
    values = s.split(" ")
    operation = values[0]
    amount = int(values[1])
    if operation=="D":
        netAmount+=amount
    elif operation=="W":
        netAmount-=amount
    else:
        pass
print(netAmount)
```

Question 18

Level 3

Question:

A website requires the users to input username and password to register. Write a program to check the validity of password input by users.

Following are the criteria for checking the password:

1. At least 1 letter between [a-z]
2. At least 1 number between [0-9]
3. At least 1 letter between [A-Z]
4. At least 1 character from [\$#@]
5. Minimum length of transaction password: 6
6. Maximum length of transaction password: 12

Your program should accept a sequence of comma separated passwords and will check them

according to the above criteria. Passwords that match the criteria are to be printed, each separated by a comma.

Example

If the following passwords are given as input to the program:

ABd1234@1,a F1#,2w3E*,2We3345

Then, the output of the program should be:

ABd1234@1

Hints:

In case of input data being supplied to the question, it should be assumed to be a console input.

Solutions:

```
import re
value = []
items=[x for x in input().split(',')]
for p in items:
    if len(p)<6 or len(p)>12:
        continue
    else:
        pass
    if not re.search("[a-z]",p):
        continue
    elif not re.search("[0-9]",p):
        continue
    elif not re.search("[A-Z]",p):
        continue
    elif not re.search("[$#@]",p):
        continue
    elif re.search("\s",p):
        continue
    else:
        pass
    value.append(p)
print(",".join(value))
```

Question 19

Level 3

Question:

You are required to write a program to sort the (name, age, height) tuples by ascending order where name is string, age and height are numbers. The tuples are input by console. The sort criteria is:

- 1: Sort based on name;
- 2: Then sort based on age;
- 3: Then sort by score.

The priority is that name > age > score.

If the following tuples are given as input to the program:

Tom,19,80

John,20,90

Jony,17,91

Jony,17,93

Json,21,85

Then, the output of the program should be:

[('John', '20', '90'), ('Jony', '17', '91'), ('Jony', '17', '93'), ('Json', '21', '85'), ('Tom', '19', '80')]

Hints:

In case of input data being supplied to the question, it should be assumed to be a console input.

We use itemgetter to enable multiple sort keys.

Solutions:

```
from operator import itemgetter, attrgetter
```

```
l = []
while True:
    s = input()
    if not s:
        break
    l.append(tuple(s.split(",")))

print(sorted(l, key=itemgetter(0,1,2)))
```

Question 20

Level 3

Question:

Define a class with a generator which can iterate the numbers, which are divisible by 7, between a given range 0 and n.

Hints:

Consider use yield

Solution:

```
def putNumbers (n) :
    i = 0
    while i<n:
        j=i
        i=i+1
        if j%7==0:
            yield j

for i in reverse(100):
    print(i)
```

Question 21

Level 3

Question

A robot moves in a plane starting from the original point (0,0). The robot can move toward UP, DOWN, LEFT and RIGHT with a given steps. The trace of robot movement is shown as the following:

UP 5

DOWN 3

LEFT 3

RIGHT 2

i

The numbers after the direction are steps. Please write a program to compute the distance from current position after a sequence of movement and original point. If the distance is a float, then just print the nearest integer.

Example:

If the following tuples are given as input to the program:

UP 5

DOWN 3

LEFT 3

RIGHT 2

Then, the output of the program should be:

2

Hints:

In case of input data being supplied to the question, it should be assumed to be a console input.

Solution:

```
import math
pos = [0,0]
```

```

while True:
    s = input()
    if not s:
        break
    movement = s.split(" ")
    direction = movement[0]
    steps = int(movement[1])
    if direction=="UP":
        pos[0]+=steps
    elif direction=="DOWN":
        pos[0]-=steps
    elif direction=="LEFT":
        pos[1]-=steps
    elif direction=="RIGHT":
        pos[1]+=steps
    else:
        pass

print(int(round(math.sqrt(pos[1]**2+pos[0]**2))))

```

Question 22

Level 3

Question:

Write a program to compute the frequency of the words from the input. The output should output after sorting the key alphanumerically.

Suppose the following input is supplied to the program:

New to Python or choosing between Python 2 and Python 3? Read Python 2 or Python 3.

Then, the output should be:

2:2

3.:1

3?:1

New:1

Python:5

Read:1

and:1

between:1

choosing:1

or:2

to:1

Hints

In case of input data being supplied to the question, it should be assumed to be a console input.

Solution:

```
freq = {}    # frequency of words in text
line = input()
for word in line.split():
    freq[word] = freq.get(word,0)+1

words = freq.keys()
words.sort()

for w in words:
    print("%s:%d" % (w,freq[w]))
```

Question 23

level 1

Question:

Write a method which can calculate square value of number

Hints:

Using the ****** operator

Solution:

```
def square(num):
    return num ** 2

print(square(2))
print(square(3))
```

Question 24

Level 1

Question:

Python has many built-in functions, and if you do not know how to use it, you can read document online or find some books. But Python has a built-in document function for every built-in functions.

Please write a program to print some Python built-in functions documents, such as `abs()`, `int()`, `raw_input()`

And add document for your own function

Hints:

The built-in document method is **doc**

Solution:

```
print(abs.__doc__)
print(int.__doc__)
print(input.__doc__)

def square(num):
    '''Return the square value of the input number.

    The input number must be integer.
    '''
    return num ** 2

print(square(2))
print(square.__doc__)
```

Question 25

Level 1

Question:

Define a class, which have a class parameter and have a same instance parameter.

Hints:

Define a instance parameter, need add it in **init** method

You can init a object with construct parameter or set the value later

Solution:

```
class Person:
    # Define the class parameter "name"
    name = "Person"

    def __init__(self, name = None):
        # self.name is the instance parameter
        self.name = name
```

```
jeffrey = Person("Jeffrey")
print("%s name is %s" % (Person.name, jeffrey.name))

nico = Person()
nico.name = "Nico"
print("%s name is %s" % (Person.name, nico.name))
```

Question 26:

Define a function which can compute the sum of two numbers.

Hints:

Define a function with two numbers as arguments. You can compute the sum in the function and return the value.

Solution

```
def SumFunction(number1, number2):
    return number1+number2

print(SumFunction(1,2))
```

Question 27

Define a function that can convert a integer into a string and print it in console.

Hints:

Use `str()` to convert a number to string.

Solution

```
def printValue(n):
    print(str(n))

printValue(3)
```

Question 28

Define a function that can convert a integer into a string and print it in console.

Hints:

Use `str()` to convert a number to string.

Solution

```
def printValue(n):  
    print(str(n))  
  
printValue(3)
```

Question 29

Define a function that can receive two integral numbers in string form and compute their sum and then print it in console.

Hints:

Use `int()` to convert a string to integer.

Solution

```
def printValue(s1,s2):  
    print(int(s1)+int(s2))  
  
printValue("3","4")
```

Question 30

Define a function that can accept two strings as input and concatenate them and then print it in console.

Hints:

Use `+` to concatenate the strings

Solution

```
def printValue(s1,s2):  
    print(s1+s2)  
  
printValue("3","4") #34
```

Question 31

Define a function that can accept two strings as input and print the string with maximum length in console. If two strings have the same length, then the function should print all strings line by line.

Hints:

Use `len()` function to get the length of a string

Solution

```
def printValue(s1,s2):
    len1 = len(s1)
    len2 = len(s2)
    if len1>len2:
        print(s1)
    elif len2>len1:
        print(s2)
    else:
        print(s1)
        print(s2)

printValue("one", "three")
```

Question 32

Define a function that can accept an integer number as input and print the "It is an even number" if the number is even, otherwise print "It is an odd number".

Hints:

Use `%` operator to check if a number is even or odd.

Solution

```
def checkValue(n):
    if n%2 == 0:
        print("It is an even number")
    else:
        print("It is an odd number")

checkValue(7)
```

Question 33

Define a function which can print a dictionary where the keys are numbers

Hints:

Use `dict[key]=value` pattern to put entry into a dictionary.

Use `**` operator to get power of a number.

Solution

```
```python
def printDict():
 d=dict()
 d[1]=1
 d[2]=2**2
 d[3]=3**2
 print(d)

printDict()
```

## Question 34

Define a function which can print a dictionary where the keys are numbers between 1 and 20 (both included) and the values are square of keys.

Hints:

Use `dict[key]=value` pattern to put entry into a dictionary.

Use `**` operator to get power of a number.

Use `range()` for loops.

Solution

```
def printDict():
 d=dict()
 for i in range(1,21):
 d[i]=i**2
 print(d)

printDict()
```

## Question 35

Define a function which can generate a dictionary where the keys are numbers between 1 and 20 (both included) and the values are square of keys. The function should just print the values only.

Hints:

Use dict[key]=value pattern to put entry into a dictionary.

Use \*\* operator to get power of a number.

Use range() for loops.

Use keys() to iterate keys in the dictionary. Also we can use item() to get key/value pairs.

Solution

```
def printDict():
 d=dict()
 for i in range(1,21):
 d[i]=i**2
 for (k,v) in d.items():
 print(v)

printDict()
```

## Question 36

Define a function which can generate a dictionary where the keys are numbers between 1 and 20 (both included) and the values are square of keys. The function should just print the keys only.

Hints:

Use dict[key]=value pattern to put entry into a dictionary.

Use \*\* operator to get power of a number.

Use range() for loops.

Use keys() to iterate keys in the dictionary. Also we can use item() to get key/value pairs.

Solution

```
def printDict():
 d=dict()
 for i in range(1,21):
 d[i]=i**2
 for k in d.keys():
 print(k)

printDict()
```

## Question 37

Define a function which can generate and print a list where the values are square of numbers between 1 and 20 (both included).

Hints:

Use `**` operator to get power of a number.

Use `range()` for loops.

Use `list.append()` to add values into a list.

Solution

```
def printList():
 li=list()
 for i in range(1,21):
 li.append(i**2)
 print(li)

printList()
```

## Question 38

Define a function which can generate a list where the values are square of numbers between 1 and 20 (both included). Then the function needs to print the first 5 elements in the list.

Hints:

Use `**` operator to get power of a number.

Use `range()` for loops.

Use `list.append()` to add values into a list.

Use `[n1:n2]` to slice a list

Solution

```
def printList():
 li=list()
 for i in range(1,21):
 li.append(i**2)
 print(li[:5])

printList()
```



## Question 39

Define a function which can generate a list where the values are square of numbers between 1 and 20 (both included). Then the function needs to print the last 5 elements in the list.

Hints:

Use `**` operator to get power of a number.

Use `range()` for loops.

Use `list.append()` to add values into a list.

Use `[n1:n2]` to slice a list

Solution

```
def printList():
 li=list()
 for i in range(1,21):
 li.append(i**2)
 print(li[-5:])

printList()
```

## Question 40

Define a function which can generate a list where the values are square of numbers between 1 and 20 (both included). Then the function needs to print all values except the first 5 elements in the list.

Hints:

Use `**` operator to get power of a number.

Use `range()` for loops.

Use `list.append()` to add values into a list.

Use `[n1:n2]` to slice a list

Solution

```
def printList():
 li=list()
 for i in range(1,21):
 li.append(i**2)
 print li[5:]

printList()
```

## Question 41

Define a function which can generate and print a tuple where the value are square of numbers between 1 and 20 (both included).

Hints:

Use **\*\*** operator to get power of a number.

Use **range()** for loops.

Use **list.append()** to add values into a list.

Use **tuple()** to get a tuple from a list.

Solution

```
def printTuple():
 li=list()
 for i in range(1,21):
 li.append(i**2)
 print(tuple(li))

printTuple()
```

## Question 42

With a given tuple (1,2,3,4,5,6,7,8,9,10), write a program to print the first half values in one line and the last half values in one line.

Hints:

Use **[n1:n2]** notation to get a slice from a tuple.

Solution

```
tp=(1,2,3,4,5,6,7,8,9,10)
tp1=tp[:5]
tp2=tp[5:]
print(tp1)
print(tp2)
```

## Question 43

Write a program to generate and print another tuple whose values are even numbers in the given tuple (1,2,3,4,5,6,7,8,9,10).

Hints:

Use "for" to iterate the tuple

Use tuple() to generate a tuple from a list.

Solution

```
tp=(1,2,3,4,5,6,7,8,9,10)
li=list()
for i in tp:
 if tp[i]%2==0:
 li.append(tp[i])

tp2=tuple(li)
print(tp2)
```

## Question 44

Write a program which accepts a string as input to print "Yes" if the string is "yes" or "YES" or "Yes", otherwise print "No".

Hints:

Use if statement to judge condition.

Solution

```
s= raw_input()
if s=="yes" or s=="YES" or s=="Yes":
 print "Yes"
else:
 print "No"
```

## Question 45

Write a program which can filter even numbers in a list by using filter function. The list is: [1,2,3,4,5,6,7,8,9,10].

Hints:

Use filter() to filter some elements in a list.

Use lambda to define anonymous functions.

Solution

```
li = [1,2,3,4,5,6,7,8,9,10]
evenNumbers = filter(lambda x: x%2==0, li)
print(evenNumbers)
```

## Question 46

Write a program which can map() to make a list whose elements are square of elements in [1,2,3,4,5,6,7,8,9,10].

Hints

Use map() to generate a list.

Use lambda to define anonymous functions.

Solution

```
li = [1,2,3,4,5,6,7,8,9,10]
squaredNumbers = map(lambda x: x**2, li)
print(squaredNumbers)
```

## Question 47

Write a program which can map() and filter() to make a list whose elements are square of even number in [1,2,3,4,5,6,7,8,9,10].

Hints

Use map() to generate a list.

Use filter() to filter elements of a list.

Use lambda to define anonymous functions.

Solution

```
li = [1,2,3,4,5,6,7,8,9,10]
evenNumbers = map(lambda x: x**2, filter(lambda x: x%2==0, li))
print(evenNumbers)
```

## Question 48

Write a program which can filter() to make a list whose elements are even number between 1 and 20 (both included).

Hints:

Use filter() to filter elements of a list.

Use lambda to define anonymous functions.

Solution

```
evenNumbers = filter(lambda x: x%2==0, range(1,21))
print(evenNumbers)
```

## Question 49

Write a program which can map() to make a list whose elements are square of numbers between 1 and 20 (both included).

Hints

Use map() to generate a list.

Use lambda to define anonymous functions.

Solution

```
squaredNumbers = map(lambda x: x**2, range(1,21))
print(squaredNumbers)
```

## Question 50

Define a class named American which has a static method called printNationality.

Hints:

Use @staticmethod decorator to define class static method.

Solution

```
class American(object):
 @staticmethod
 def printNationality():
 print("America")

anAmerican = American()
```

```
anAmerican.printNationality()
American.printNationality()
```

## Question 51

Define a class named American and its subclass NewYorker.

Hints:

Use class Subclass(ParentClass) to define a subclass.

Solution:

```
class American(object):
 pass

class NewYorker(American):
 pass

anAmerican = American()
aNewYorker = NewYorker()
print(anAmerican)
print(aNewYorker)
```

## Question 52

Define a class named Circle which can be constructed by a radius. The Circle class has a method which can compute the area.

Hints:

Use def methodName(self) to define a method.

Solution:

```
class Circle(object):
 def __init__(self, r):
 self.radius = r

 def area(self):
 return self.radius**2*3.14
```

```
aCircle = Circle(2)
print aCircle.area()
```

## Question 53

Define a class named Rectangle which can be constructed by a length and width. The Rectangle class has a method which can compute the area.

Hints:

Use `def methodName(self)` to define a method.

Solution:

```
class Rectangle(object):
 def __init__(self, l, w):
 self.length = l
 self.width = w

 def area(self):
 return self.length*self.width

aRectangle = Rectangle(2,10)
print(aRectangle.area())
```

## Question 54

Define a class named Shape and its subclass Square. The Square class has an init function which takes a length as argument. Both classes have a area function which can print the area of the shape where Shape's area is 0 by default.

Hints:

To override a method in super class, we can define a method with the same name in the super class.

Solution:

```
class Shape(object):
 def __init__(self):
 pass

 def area(self):
 return 0
```

```

class Square(Shape):
 def __init__(self, l):
 Shape.__init__(self)
 self.length = l

 def area(self):
 return self.length*self.length

aSquare= Square(3)
print(aSquare.area())

```

## Question 55

Please raise a RuntimeError exception.

Hints:

Use raise() to raise an exception.

Solution:

```

raise RuntimeError('something wrong')

```

## Question 56

Write a function to compute 5/0 and use try/except to catch the exceptions.

Hints:

Use try/except to catch exceptions.

Solution:

```

def throws():
 return 5/0

try:
 throws()
except ZeroDivisionError:
 print("division by zero!")
except Exception, err:
 print('Caught an exception')

```



```
finally:
 print('In finally block for cleanup')
```

## Question 57

Define a custom exception class which takes a string message as attribute.

Hints:

To define a custom exception, we need to define a class inherited from Exception.

Solution:

```
class MyError(Exception):
 """My own exception class

 Attributes:
 msg -- explanation of the error
 """

 def __init__(self, msg):
 self.msg = msg

error = MyError("something wrong")
```

## Question 58

Assuming that we have some email addresses in the "[username@companyname.com](#)" format, please write program to print the user name of a given email address. Both user names and company names are composed of letters only.

Example:

If the following email address is given as input to the program:

[john@google.com](mailto:john@google.com)

Then, the output of the program should be:

john

In case of input data being supplied to the question, it should be assumed to be a console input.

Hints:

Use \w to match letters.

Solution:

```
import re
emailAddress = raw_input()
pat2 = "(\w+)@((\w+\.)+(com))"
r2 = re.match(pat2,emailAddress)
print(r2.group(1))
```

## Question 59

Assuming that we have some email addresses in the "[username@companyname.com](#)" format, please write program to print the company name of a given email address. Both user names and company names are composed of letters only.

Example:

If the following email address is given as input to the program:

[john@google.com](#)

Then, the output of the program should be:

google

In case of input data being supplied to the question, it should be assumed to be a console input.

Hints:

Use \w to match letters.

Solution:

```
import re
emailAddress = raw_input()
pat2 = "(\w+)@(\w+)\.(com)"
r2 = re.match(pat2,emailAddress)
print(r2.group(2))
```

## Question 60

Write a program which accepts a sequence of words separated by whitespace as input to print the words composed of digits only.

Example:

If the following words is given as input to the program:

2 cats and 3 dogs.

Then, the output of the program should be:

['2', '3']

In case of input data being supplied to the question, it should be assumed to be a console input.

Hints:

Use `re.findall()` to find all substring using regex.

Solution:

```
import re
s = raw_input()
print(re.findall("\d+",s))
```

## Question 61

Print a unicode string "hello world".

Hints:

Use u'strings' format to define unicode string.

Solution:

```
unicodeString = u"hello world!"
print(unicodeString)
```

## Question 62

Write a program to read an ASCII string and to convert it to a unicode string encoded by utf-8.

Hints:

Use `unicode()` function to convert.

Solution:

```
s = input()
u = unicode(s , "utf-8")
print(u)
```

## Question 63

Write a special comment to indicate a Python source code file is in unicode.

Hints:

Solution:

```
-*- coding: utf-8 -*-

#-----#
```

## Question 64

Write a program to compute  $1/2+2/3+3/4+\dots+n/n+1$  with a given  $n$  input by console ( $n>0$ ).

Example:

If the following  $n$  is given as input to the program:

5

Then, the output of the program should be:

3.55

In case of input data being supplied to the question, it should be assumed to be a console input.

Hints:

Use `float()` to convert an integer to a float

Solution:

```
n=int(input())
sum=0.0
for i in range(1,n+1):
 sum += float(float(i)/(i+1))
print(sum)
```

## Question 65

Write a program to compute:

$f(n)=f(n-1)+100$  when  $n>0$

and  $f(0)=1$

with a given  $n$  input by console ( $n>0$ ).

Example:

If the following  $n$  is given as input to the program:

5

Then, the output of the program should be:

500

In case of input data being supplied to the question, it should be assumed to be a console input.

Hints:

We can define recursive function in Python.

Solution:

```
def f(n):
 if n==0:
 return 0
 else:
 return f(n-1)+100
```

```
n=int(input())
print(f(n))
```

## Question 66

The Fibonacci Sequence is computed based on the following formula:

$f(n)=0$  if  $n=0$

$f(n)=1$  if  $n=1$

$f(n)=f(n-1)+f(n-2)$  if  $n>1$

Please write a program to compute the value of  $f(n)$  with a given  $n$  input by console.

Example:

If the following n is given as input to the program:

7

Then, the output of the program should be:

13

In case of input data being supplied to the question, it should be assumed to be a console input.

Hints:

We can define recursive function in Python.

Solution:

```
def f(n):
 if n == 0: return 0
 elif n == 1: return 1
 else: return f(n-1)+f(n-2)

n=int(input())
print(f(n))
```

## Question 67

The Fibonacci Sequence is computed based on the following formula:

$f(n)=0$  if  $n=0$

$f(n)=1$  if  $n=1$

$f(n)=f(n-1)+f(n-2)$  if  $n>1$

Please write a program using list comprehension to print the Fibonacci Sequence in comma separated form with a given n input by console.

Example:

If the following n is given as input to the program:

7

Then, the output of the program should be:

0,1,1,2,3,5,8,13

Hints:

We can define recursive function in Python.

Use list comprehension to generate a list from an existing list.

Use string.join() to join a list of strings.

In case of input data being supplied to the question, it should be assumed to be a console input.

Solution:

```
def f(n):
 if n == 0: return 0
 elif n == 1: return 1
 else: return f(n-1)+f(n-2)

n=int(input())
values = [str(f(x)) for x in range(0, n+1)]
print(",".join(values))
```

## Question 68

Please write a program using generator to print the even numbers between 0 and n in comma separated form while n is input by console.

Example:

If the following n is given as input to the program:

10

Then, the output of the program should be:

0,2,4,6,8,10

Hints:

Use yield to produce the next value in generator.

In case of input data being supplied to the question, it should be assumed to be a console input.

Solution:

```
def EvenGenerator(n):
 i=0
 while i<=n:
 if i%2==0:
 yield i
 i+=1
```

```

n=int(input())
values = []
for i in EvenGenerator(n):
 values.append(str(i))

print(",".join(values))

```

## Question 69

Please write a program using generator to print the numbers which can be divisible by 5 and 7 between 0 and n in comma separated form while n is input by console.

Example:

If the following n is given as input to the program:

100

Then, the output of the program should be:

0,35,70

Hints:

Use yield to produce the next value in generator.

In case of input data being supplied to the question, it should be assumed to be a console input.

Solution:

```

def NumGenerator(n):
 for i in range(n+1):
 if i%5==0 and i%7==0:
 yield i

n=int(input())
values = []
for i in NumGenerator(n):
 values.append(str(i))

print(",".join(values))

```

## Question 70



Please write assert statements to verify that every number in the list [2,4,6,8] is even.

Hints:

Use "assert expression" to make assertion.

Solution:

```
li = [2,4,6,8]
for i in li:
 assert i%2==0
```

## Question 71

Please write a program which accepts basic mathematic expression from console and print the evaluation result.

Example:

If the following string is given as input to the program:

35+3

Then, the output of the program should be:

38

Hints:

Use eval() to evaluate an expression.

Solution:

```
expression = raw_input()
print(eval(expression))
```

## Question 72

Please write a binary search function which searches an item in a sorted list. The function should return the index of element to be searched in the list.

Hints:

Use if/elif to deal with conditions.

Solution:

```

import math
def bin_search(li, element):
 bottom = 0
 top = len(li)-1
 index = -1
 while top>=bottom and index==-1:
 mid = int(math.floor((top+bottom)/2.0))
 if li[mid]==element:
 index = mid
 elif li[mid]>element:
 top = mid-1
 else:
 bottom = mid+1

 return index

li=[2,5,7,9,11,17,222]
print(bin_search(li,11))
print(bin_search(li,12))

```

## Question 73

Please write a binary search function which searches an item in a sorted list. The function should return the index of element to be searched in the list.

Hints:

Use if/elif to deal with conditions.

Solution:

```

import math
def bin_search(li, element):
 bottom = 0
 top = len(li)-1
 index = -1
 while top>=bottom and index==-1:
 mid = int(math.floor((top+bottom)/2.0))
 if li[mid]==element:
 index = mid
 elif li[mid]>element:
 top = mid-1
 else:

```

```
 bottom = mid+1

 return index

li=[2,5,7,9,11,17,222]
print(bin_search(li,11))
print(bin_search(li,12))
```

## Question 74

Please generate a random float where the value is between 10 and 100 using Python math module.

Hints:

Use `random.random()` to generate a random float in `[0,1]`.

Solution:

```
import random
print(random.random()*100)
```

## Question 75

Please generate a random float where the value is between 5 and 95 using Python math module.

Hints:

Use `random.random()` to generate a random float in `[0,1]`.

Solution:

```
import random
print(random.random()*100-5)
```

## Question 76

Please write a program to output a random even number between 0 and 10 inclusive using random module and list comprehension.

Hints:

Use `random.choice()` to a random element from a list.

Solution:

```
import random
print(random.choice([i for i in range(11) if i%2==0]))
```

## Question 77

Please write a program to output a random number, which is divisible by 5 and 7, between 0 and 10 inclusive using random module and list comprehension.

Hints:

Use random.choice() to a random element from a list.

Solution:

```
import random
print(random.choice([i for i in range(201) if i%5==0 and i%7==0]))
```

## Question 78

Please write a program to generate a list with 5 random numbers between 100 and 200 inclusive.

Hints:

Use random.sample() to generate a list of random values.

Solution:

```
import random
print(random.sample(range(100), 5))
```

## Question 79

Please write a program to randomly generate a list with 5 even numbers between 100 and 200 inclusive.

Hints:

Use random.sample() to generate a list of random values.

Solution:

```
import random
print(random.sample([i for i in range(100,201) if i%2==0], 5))
```

## Question 80

Please write a program to randomly generate a list with 5 numbers, which are divisible by 5 and 7 , between 1 and 1000 inclusive.

Hints:

Use `random.sample()` to generate a list of random values.

Solution:

```
import random
print(random.sample([i for i in range(1,1001) if i%5==0 and i%7==0], 5))
```

## Question 81

Please write a program to randomly print a integer number between 7 and 15 inclusive.

Hints:

Use `random.randrange()` to a random integer in a given range.

Solution:

```
import random
print(random.randrange(7,16))
```

## Question 82

Please write a program to compress and decompress the string "hello world!hello world!hello world!hello world!".

Hints:

Use `zlib.compress()` and `zlib.decompress()` to compress and decompress a string.

Solution:

```
import zlib
s = b'hello world!hello world!hello world!hello world!'
t = zlib.compress(s)
print(t)
print(zlib.decompress(t))
```

## Question 83

Please write a program to print the running time of execution of "1+1" for 100 times.

Hints:

Use `timeit()` function to measure the running time.

Solution:

```
from timeit import Timer
t = Timer("for i in range(100):1+1")
print(t.timeit())
```

## Question 84

Please write a program to shuffle and print the list [3,6,7,8].

Hints:

Use `shuffle()` function to shuffle a list.

Solution:

```
from random import shuffle
li = [3,6,7,8]
shuffle(li)
print(li)
```

## Question 85

Please write a program to shuffle and print the list [3,6,7,8].

Hints:

Use `shuffle()` function to shuffle a list.

Solution:

```
from random import shuffle
li = [3,6,7,8]
shuffle(li)
print(li)
```

## Question 86

Please write a program to generate all sentences where subject is in ["I", "You"] and verb is in ["Play", "Love"] and the object is in ["Hockey", "Football"].

Hints:

Use list[index] notation to get a element from a list.

Solution:

```
subjects=["I", "You"]
verbs=["Play", "Love"]
objects=["Hockey","Football"]
for i in range(len(subjects)):
 for j in range(len(verbs)):
 for k in range(len(objects)):
 sentence = "%s %s %s." % (subjects[i], verbs[j], objects[k])
 print(sentence)
```

## Question 87

Please write a program to print the list after removing delete even numbers in [5,6,77,45,22,12,24].

Hints:

Use list comprehension to delete a bunch of element from a list.

Solution:

```
li = [5,6,77,45,22,12,24]
li = [x for x in li if x%2!=0]
print(li)
```

## Question 88

By using list comprehension, please write a program to print the list after removing delete numbers which are divisible by 5 and 7 in [12,24,35,70,88,120,155].

Hints:

Use list comprehension to delete a bunch of element from a list.

Solution:

```
li = [12,24,35,70,88,120,155]
li = [x for x in li if x%5!=0 and x%7!=0]
print(li)
```

## Question 89

By using list comprehension, please write a program to print the list after removing the 0th, 2nd, 4th, 6th numbers in [12,24,35,70,88,120,155].

Hints:

Use list comprehension to delete a bunch of element from a list.

Use enumerate() to get (index, value) tuple.

Solution:

```
li = [12,24,35,70,88,120,155]
li = [x for (i,x) in enumerate(li) if i%2!=0]
print(li)
```

## Question 90

By using list comprehension, please write a program generate a 3x8 2D array whose each element is 0.

Hints:

Use list comprehension to make an array.

Solution:

```
array = [[[0 for col in range(8)] for col in range(5)] for row in
range(3)]
print(array)
```

## Question 91

By using list comprehension, please write a program to print the list after removing the 0th, 4th, 5th numbers in [12,24,35,70,88,120,155].

Hints:

Use list comprehension to delete a bunch of element from a list.

Use enumerate() to get (index, value) tuple.

Solution:



```
li = [12,24,35,70,88,120,155]
li = [x for (i,x) in enumerate(li) if i not in (0,4,5)]
print(li)
```

## Question 92

By using list comprehension, please write a program to print the list after removing the value 24 in [12,24,35,24,88,120,155].

Hints:

Use list's remove method to delete a value.

Solution:

```
li = [12,24,35,24,88,120,155]
li = [x for x in li if x!=24]
print(li)
```

## Question 93

With two given lists [1,3,6,78,35,55] and [12,24,35,24,88,120,155], write a program to make a list whose elements are intersection of the above given lists.

Hints:

Use set() and "&=" to do set intersection operation.

Solution:

```
set1=set([1,3,6,78,35,55])
set2=set([12,24,35,24,88,120,155])
set1 &= set2
li=list(set1)
print(li)
```

## Question 94

With a given list [12,24,35,24,88,120,155,88,120,155], write a program to print this list after removing all duplicate values with original order reserved.

Hints:

Use set() to store a number of values without duplicate.

Solution:

```
def removeDuplicate(li):
 newli=[]
 seen = set()
 for item in li:
 if item not in seen:
 seen.add(item)
 newli.append(item)

 return newli

li=[12,24,35,24,88,120,155,88,120,155]
print(removeDuplicate(li))
```

## Question 95

Define a class Person and its two child classes: Male and Female. All classes have a method "getGender" which can print "Male" for Male class and "Female" for Female class.

Hints:

Use Subclass(Parentclass) to define a child class.

Solution:

```
class Person(object):
 def getGender(self):
 return "Unknown"

class Male(Person):
 def getGender(self):
 return "Male"

class Female(Person):
 def getGender(self):
 return "Female"

aMale = Male()
aFemale= Female()
```

```
print(aMale.getGender())
print(aFemale.getGender())
```

## Question 96

Please write a program which count and print the numbers of each character in a string input by console.

Example:

If the following string is given as input to the program:

abcdefgabc

Then, the output of the program should be:

a,2  
c,2  
b,2  
e,1  
d,1  
g,1  
f,1

Hints:

Use dict to store key/value pairs.

Use dict.get() method to lookup a key with default value.

Solution:

```
dic = {}
s=raw_input()
for s in s:
 dic[s] = dic.get(s,0)+1
print('\n'.join(['%s,%s' % (k, v) for k, v in dic.items()])))
```

## Question 97

Please write a program which accepts a string from console and print it in reverse order.

Example:

If the following string is given as input to the program:

rise to vote sir

Then, the output of the program should be:

ris etov ot esir

Hints:

Use list[::-1] to iterate a list in a reverse order.

Solution:

```
s=raw_input()
s = s[::-1]
print(s)
```

## Question 98

Please write a program which accepts a string from console and print the characters that have even indexes.

Example:

If the following string is given as input to the program:

H1e2l3l4o5w6o7r8l9d

Then, the output of the program should be:

Helloworld

Hints:

Use list[::2] to iterate a list by step 2.

Solution:

```
s=raw_input()
s = s[::2]
print(s)
```

## Question 99

Please write a program which prints all permutations of [1,2,3]

Hints:

Use itertools.permutations() to get permutations of list.

Solution:

```
import itertools
print(list(itertools.permutations([1,2,3])))
```

## Question 100

Write a program to solve a classic ancient Chinese puzzle:

We count 35 heads and 94 legs among the chickens and rabbits in a farm. How many rabbits and how many chickens do we have?

Hint:

Use for loop to iterate all possible solutions.

Solution:

```
def solve(numheads,numlegs):
 ns='No solutions!'
 for i in range(numheads+1):
 j=numheads-i
 if 2*i+4*j==numlegs:
 return i,j
 return ns,ns

numheads=35
numlegs=94
solutions=solve(numheads,numlegs)
print(solutions)
```